# DC Microstep Drive with Q Programming (ST5-Q-NN)

Selection data & performance | Manufacturer technical reference.

## Product Features

### FEATURES

- Programmable microstepping drive with advanced current control
- Q Programmer™ for robust motion control programming
- 744 lines of stored program capability
- Math calculations using analog and digital parameters
- Supports all "S" drive control modes as well
- Modbus/RTU for easy PLC and HMI communication
- Compatible with many 3rd party HMIs
- Wide current range 0.1 to 5.0 A/phase (peak of sine) with idle current reduction
- Advanced anti-resonance algorithm
- Torque ripple smoothing
- 8 digital inputs, 4 digital outputs, optically isolated
- 2 analog inputs, +/-10 volt range
- RS-232 cable and mating connectors included



### DESCRIPTION

The ST5-Q-NN stepper drive is a DC-powered microstepping drive for controlling two-phase, bipolar step motors. It offers advanced current control and a sophisticated 3rd generation anti-resonance algorithm that electronically dampens motor and system resonances to improve motor smoothness and usable torque over a wide speed range. The drive also employs electronic torque ripple smoothing and microstep emulation to greatly reduce motor noise and vibration. The drive must be powered from 24-48 VDC and can output up to 5.0 A/phase (peak-of-sine) to the step motor. Over-voltage, over-temperature and over-current protection features prevent damage while running in adverse conditions. The drive is complemented by a specifically matched set of NEMA 11 through NEMA 23 frame stepper motors (see Related and Recommended products below).

The ST5-Q-NN can operate in all of the same control modes as the S model (pulse & direction, velocity, streaming commands, SiNet Hub operation), plus it has the ability to run stand-alone Q programs stored in non-volatile memory. Q programs are created using the Q Programmer™ software, which provides multi-tasking, math functions, conditional processing, data register manipulation, and more features in a robust yet simple text-based programming language. The drive is setup and configured using Applied Motion's ST Configurator™ software. Preconfigured motor setup files included with ST Configurator™ make it easy to set up the drive for optimum results.

For connecting to external devices such as control signals, incremental encoders, limit switches, proximity or photoelectric sensors, PLC I/O, lamps, and other devices, the drive comes with 8 digital inputs, 4 digital outputs, and 2 single-ended analog inputs (analog inputs can be wired together as 1 differential analog input). Adjustable digital filters are present on the digital inputs for enhanced reliability in noisy environments.

The drive comes with an RS-232 port for configuration, serial communications and programming.

All ST drives are CE approved and RoHS compliant.

Match motor current, voltage, command interface and operating profile. Specify wiring, enclosure and interlocks; this is an electrical accessory, not a fluid valve. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

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Dimensions, connections & configuration | Manufacturer technical reference.

| SPECIFICATIONS | |
|---|---|
| Model Number | ST5-Q-NN |
| Part Number | 5000-128 |
| Supply Voltage | 24-48 VDC |
| Supply Voltage Type | DC |
| Control Modes | · Step & Direction<br>· Velocity (Oscillator)<br>· Streaming Commands<br>· Analog Positioning<br>· Encoder Following<br>· Q Programming<br>· SiNet Hub Compatible<br>· Modbus |
| Output Current | 0.1-5.0 A/Phase |
| Communication Ports | RS-232 |
| Encoder Feedback | No |
| Step Resolution | · Full<br>· Half<br>· Microstepping<br>· Microstep Emulation |
| Idle Current Reduction | 0-90% |
| Setup Method | Software setup |
| Digital Inputs | 8 |
| Digital Outputs | 4 |
| Analog Inputs | 1 differential or 2 single-ended |
| Dimensions | 5.0 x 3.0 x 1.75 inches |
| Weight | 10.4 oz |
| Operating Temperature Range | 0-70 °C |
| Ambient Temperature Range | 0-55 °C |
| Ambient Humidity | 90% max, non-condensing |
| Status LEDs | 1 red, 1 green |
| Circuit Protection<br>3/2 Valve Spring<br>to Open P | · Short circuit<br>· Over-voltage<br>· Under-voltage<br>· Over-temp |

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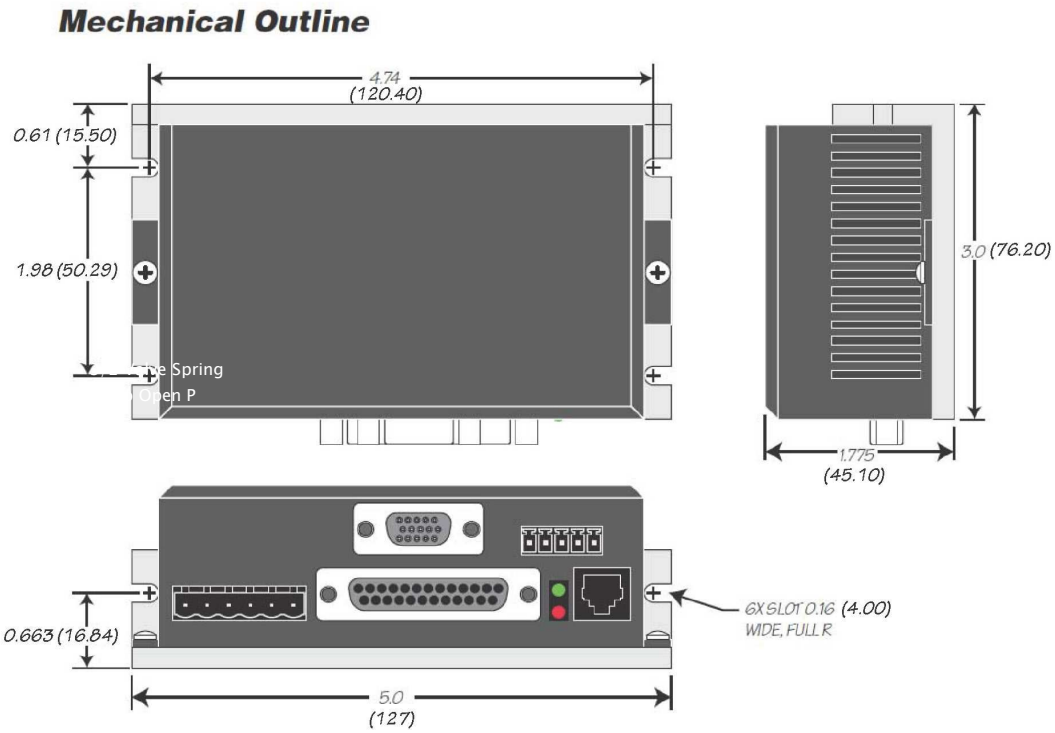
# DC Microstep Drive with Q Programming (ST5-Q-NN)

Dimensions, connections & configuration | Manufacturer technical reference.

| DOWNLOADS & SOFTWARE | |
|---|---|
| Manuals | ST5 10-Q QuickSetup 920-0007E.pdf<br>Host Command Reference Rev I.pdf<br>ST5-10-QSi Hardware Manual 920-0004F.pdf<br>ModbusManual 920-0072.pdf |
| Datasheet | ST Datasheet 925-0007.pdf |
| 2D Drawing | ST5 10 Dimensions.pdf<br>ST T simple 3D.pdf |
| 3D Drawing | ST5 10-Q Si C SIMPLE.igs |
| Speed-Torque Curves | ST speed-torque.pdf |
| Agency Approvals | ST-Q-Si-C-IP CE DOC.pdf |
| Application Notes | APPN0021 5V-Keepalive-Circuit.pdf<br>APPN0019 Analog-positioning-using-Q-program.zip<br>APPN0018_EZ-Series-Touchpanel-HMI.zip<br>APPN0016_Simple-25-pin-mating-connections.pdf<br>APPN0015_Make-a-serial-programming-cable.pdf |
| Software Downloads | SCL Utility<br>ST Configurator™ |
| Sample Code | scldemo.zip |

**All Downloads and Software can be downloaded from TWHC website**

## 2D DRAWING




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# DC Microstep Drive with Q Programming (ST5-Q-NN)

Dimensions, connections & configuration | Manufacturer technical reference.

## PRODUCTS IN THE SERIES *MODBUS PRODUCTS*

| MODEL NUMBER | SUPPLY VOLTAGE | CONTROL MODES | OUTPUT CURRENT | COMMUNICATION PORTS |
|---|---|---|---|---|
| ST5-Q-NN | 24-48 VDC | Step & Direction<br>Velocity (Oscillator)<br>Streaming Commands<br>Analog Positioning<br>Encoder Following<br>Q Programming<br>SiNet Hub Compatible<br>Modbus | 0.1-5.0 A/Phase | RS-232 |
| ST5-Q-NN | 24-48 VDC | Step & Direction<br>Velocity (Oscillator)<br>Streaming Commands<br>Analog Positioning<br>Encoder Following<br>Q Programming<br>Modbus | 0.1-5.0 A/Phase | RS-232<br>RS-485 |

## PRODUCTS IN THE SERIES *ST STEPPER DRIVES*

| MODEL NUMBER | SUPPLY VOLTAGE | CONTROL MODES | OUTPUT CURRENT | COMMUNICATION PORTS |
|---|---|---|---|---|
| ST5-Q-NN | 24-48 VDC | Step & Direction<br>Velocity (Oscillator)<br>Streaming Commands<br>Analog Positioning<br>Encoder Following<br>Q Programming<br>SiNet Hub Compatible<br>Modbus | 0.1-5.0 A/Phase | RS-232 |

***For Integrated encoder feedback consult TWHC**

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| PART NUMBER | DESCRIPTION |
|---|---|
| 561-002 | Open Loop Flow Control Joystick Demand |
| 561-003 | Closed Loop Pressure Control Digital Demand |
| 561-004 | Closed Loop Pressure Control Analogue Demand 0-10v DC |
| 561-005 | Open Loop Flow Control Analogue Demand Uni-Directional 0-10v DC |
| 561-006 | Open Loop Flow Control Analogue Demand Bi-Directional 0-10v DC |

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